Group Projects Proposals

30562 - MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

General Recommendations

- A good strategy is often to start from one or more of the attached papers (or related ones), understand the setup, and reproduce a small part of the reported results before extending the study in your own direction.
- Creativity is strongly encouraged, but you are absolutely **not** expected to beat any state-of-the-art benchmark. For this course, a clear question, a careful experimental design, and an original small-scale analysis are more valuable than an overly ambitious project.
- **Keep the scale of the experiments realistic**. For LLM projects, do not expect to fine-tune models much larger than roughly 1B parameters, and for inference you should stay at or usually below about 7B parameters.
- Take hardware and time constraints seriously from the beginning. Start with a small baseline, verify that the pipeline works end to end, and only then scale up if the setup remains manageable.
- It is usually better to answer one focused question well than to run many loosely connected experiments. Try to define early what you want to measure, what a fair comparison looks like, and what result would count as informative.

Project Proposals

1. **Ablation Study of ResNets or U-Nets** *Structured*
   Perform a controlled ablation study of the main components of a standard architecture, such as a ResNet for image classification or a U-Net for image segmentation. Possible directions include removing skip connections, changing normalization, varying depth, modifying block structure, or testing how encoder-decoder connections affect performance, provided that the setup stays simple and the comparison is fair [1, 2].

2. **Do MLPs Learn Convolution-Like Structure on Images?** *Structured*
   Study whether a multilayer perceptron trained on image data can acquire internal structure or behavior resembling that of convolutional models, and whether data augmentation encourages this effect. Possible directions include comparing an MLP with a small CNN on the same task, visualizing learned weights, measuring sensitivity to translations, and testing whether augmentation makes the learned representations more local, reusable, or spatially regular [3, 4, 5, 6].

3. **Autoencoders, VAEs, and the Geometry of Latent Spaces** *Structured*
   Explore what kind of latent space is learned by an autoencoder or VAE. You may focus on interpolation, disentanglement, prior choice, or interpretability, as long as the experiments help explain what the latent representation captures [7, 8, 9].

4. **Full Fine-Tuning vs Parameter-Efficient Adaptation** *Structured*
   Compare full fine-tuning with lighter adaptation methods on a task of your choice, either in language or in vision. Possible strategies include methods such as LoRA or updating only a small part of the model, for example the last layer of a ResNet. The main goal is to understand the trade-off between performance, cost, and which parts of the model are worth updating [10, 11, 12, 13].

5. **Does Curriculum Learning Help?** *Structured*
   Investigate whether the order in which training examples are presented affects learning. You are free to define what makes an example easy or hard, provided the comparison is controlled and the training budget is fair [14, 15, 16].

6. **Bias in Word Embeddings** *Structured*
   Study whether word embeddings such as word2vec or GloVe capture biases present in the data they are trained on. You may use pretrained embeddings or train your own on a manageable corpus, and analyze how associations between words change with the dataset, the training setup, or the evaluation procedure, for example through nearest neighbors, analogies, or WEAT-style tests [17, 18, 19].

7. **Transfer Learning under Controlled Domain Shift** *Intermediate*
   Study how a pre-trained model behaves when adapted to data that differ from its original domain. Choose a source-target setup that is different but still comparable, for example related datasets or a dataset with controlled transformations, and analyze what transfers well, what breaks, and which adaptation choices matter most [10, 20, 21].

8. **Can Data Augmentation Replace Hard Inductive Bias?** *Intermediate*
   Study whether stronger augmentation can compensate for weaker architectural bias. You may vary model family, augmentation strategy, dataset size, or distribution shift, and ask when augmentation changes the conclusion [4, 22, 23].

9. **How Small Can a Vision Model Be?** *Intermediate*
   Try to reach a useful level of vision performance with very limited parameters, memory, or inference cost. For example, you might ask how small a model can be while still solving an image classification task such as CIFAR-10 reasonably well on a resource-constrained device. The project is intentionally creative: define a constraint, explore ways to meet it, and justify the trade-offs you make [24, 25, 26].

10. **Characterizing Distillation** *Intermediate*
    Study what is gained or lost when a smaller model learns from a larger one. You can compare distillation methods, teacher-student gaps, or evaluation criteria, and ask what kind of knowledge is actually transferred [27, 28, 29].

11. **Catastrophic Forgetting after Fine-Tuning** *Intermediate*
Study what a model forgets when it is fine-tuned sequentially on multiple tasks or domains. You can compare task pairs, adaptation methods, or mitigation strategies, as long as forgetting is measured clearly [30, 31, 32].

12. **Optimizers, Sharpness, and Generalization** *Intermediate*
Compare a small set of optimizers and study how the solutions they find differ. You may connect performance with sharpness, flatness, training dynamics, robustness, or tuning sensitivity [33, 34, 35, 36, 37, 38, 39].

13. **CNNs vs Vision Transformers under Data Scarcity** *Intermediate*
Compare a small CNN and a small Vision Transformer when data are limited. Design the experiment so as to probe the role of the different inductive biases of the two architectures, which should be especially visible in the low-data regime, while keeping the comparison fair and informative [3, 22, 23].

14. **Reasoning Gym Mini-Lab** *Intermediate*
Reasoning Gym is a benchmark of procedurally generated reasoning tasks, which makes it easy to create large amounts of synthetic data for controlled experiments. Select one or a few tasks, study how a small language model performs on them, and either try to improve it through methods such as fine-tuning or focus on a careful analysis of its strengths and failures [40, 41].

15. **When Does Retrieval-Augmented Generation Help?** *Intermediate*
Build a small retrieval-augmented system on a corpus of your choice and investigate how much different parts of the pipeline affect the final results. Possible directions include chunk size, embedding model, retriever quality, prompting, and the interaction between retrieval and generation, with the goal of understanding when retrieval helps and when it hurts [42, 43, 44].

16. **Needle-in-a-Haystack and "Lost in the Middle"** *Intermediate*
Investigate how models use relevant information placed at different positions in a long input. You may vary context length, distractors, retrieval support, or prompt design to understand when the model loses track of the important part [45, 46, 47].

17. **Lottery Tickets investigation** *Exploratory*
The lottery ticket hypothesis suggests that a large dense network may contain much smaller sparse subnetworks that can be trained to perform surprisingly well. Study whether sparse subnetworks can match the behavior of a dense model, and under what conditions. You may focus on reproducibility, sparsity level, transfer across tasks, or what properties make a ticket "good" [48, 49, 50].

18. **State-Space Models, RNNs, and Transformers on Simple Sequence Tasks** *Exploratory*
Compare a small state-space model, a standard recurrent model such as an RNN or LSTM, and a small transformer on a simple sequence problem. Beyond accuracy, you can study scaling with sequence length, memory, speed, or the types of patterns each model handles well [51, 52, 53, 54, 55].

19. Neural ODEs: Continuous-Depth Models vs Discrete Networks *Exploratory*

Neural ODEs treat a deep model as a continuous dynamical system rather than as a finite stack of layers. The goal of this project is to understand what Neural ODEs are, what their limitations are, and whether simple refinements or variants can improve them on a small problem [56, 57, 58].

20. Tokenization Matters *Exploratory*

Tokenization is often treated as a preprocessing detail, but it is in fact a crucial part of how a language model represents and processes text. Study how different tokenization choices affect the behavior and efficiency of a small language model that you can control and understand. Possible directions include rare words, multilingual text, code, arithmetic, sequence length, or robustness to noise [59, 60, 61].

21. In-Context Learning in Controlled Synthetic Tasks *Exploratory*

Use synthetic tasks to study what a transformer can learn from examples placed in its context. You can choose the task family and the variables to manipulate, but the setup should stay simple enough that the behavior can be interpreted [62, 63, 64].

22. User Modeling and Personalization in Multi-Turn Dialogue *Exploratory*

Study how well a language model can infer stable user preferences, traits, or interaction patterns from conversation history, and whether this improves its ability to predict or personalize future responses. Possible directions include evaluating held-out user turns with log-likelihood, comparing raw dialogue history with summaries or retrieved memories, and testing whether an explicit user representation extracted from past interactions helps personalization while remaining robust and privacy-aware [65, 66, 67, 68, 69].

23. Parameter Golf Challenge for Language Models *Exploratory*

Try to reach a useful level of language performance with very limited parameters, memory, or inference cost. This project is inspired by OpenAI's official Parameter Golf challenge, but you are not expected to compete under the exact same constraints. Instead, choose a reasonable set of constraints, make sure to adhere to them consistently, and keep the spirit of the challenge while exploring how architecture or adaptation choices change the trade-off [70].

24. Multimodal Reliance and Modality Shortcuts *Exploratory*

Study how a model uses image and text when both are available. You can perturb, remove, or conflict the modalities to understand when multimodal learning is genuine and when the model relies on shortcuts [71, 72, 73].

25. Beyond Standard Attention *Exploratory*

Compare standard attention with at least one alternative sequence mechanism. You may investigate quite different ideas, including attention without softmax, subquadratic mechanisms, or even superquadratic ones, as long as the comparison is meaningful. Choose a manageable task and ask what is gained or lost in efficiency and in the kinds of dependencies the model can capture [52, 74, 75, 76, 77, 78].

26. Routing and Specialisation in Toy Mixture-of-Experts Models *Exploratory*

Build a small mixture-of-experts model and study how experts specialize, if they do at all. You may focus on routing behavior, stability, load balancing, or the effect of task structure [79, 80, 81].

27. **Steering Vectors and Activation-Space Control in Language Models** *Exploratory*
Steering vectors are directions in a model's internal activation space that can sometimes be used to change its behavior. Investigate whether internal activations can be used to steer model behavior. You may study stability, side effects, layer choice, or comparison with prompt-based control [82, 83, 84, 85].

28. **Limits of Next-Token Prediction** *Exploratory*
Compare standard next-token prediction with an alternative training objective on a manageable task. The aim is to understand which strengths and weaknesses come from the autoregressive objective itself [86, 87, 88].

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